

COURSE MATERIAL

SUBJECT	ENGINEERING PHYSICS (PH23ABS101)
UNIT	III
COURSE	B.TECH
DEPARTMENT	S&H
SEMESTER	II
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1. CourseObjectives

The objective of this course are to:

Bridge the gap between the Physics in school at 10+2 level and UG level engineering courses by identifying the importance of the optical phenomenon like interference, diffraction etc, enlightening the periodic arrangement of atoms in crystalline solids and concepts of quantum mechanics, introduce novel concepts of dielectric and magnetic materials, physics of semiconductors.

1. Prerequisites

Students should have knowledge on

1. WaveOptics
2. Crystallography
3. QuantumMechanics
4. Semiconductors

2. Syllabus

Unit-III:DielectricandMagneticMaterials

Dielectric Materials: Introduction - Dielectric polarization - Dielectric polarizability, Susceptibility, Dielectric constant and Displacement Vector – Relation between the electric vectors - Types of polarizations- Electronic (Quantitative), Ionic (Quantitative) and Orientation polarizations (Qualitative)

- Lorentz internal field - Clausius- Mossotti equation – Frequency dependence of polarization – dielectric loss.

Magnetic Materials: Introduction - Magnetic dipole moment - Magnetization- Magnetic susceptibility and permeability – Atomic origin of magnetism - Classification of magnetic materials: Dia, para, Ferro, anti-ferro & Ferri magnetic materials - Domain concept for Ferromagnetism & Domain walls (Qualitative) - Hysteresis - soft and hard magnetic materials- Applications of magnetic materials.

3. Courseoutcomes

1. CO 1: Understand the intensity variation of light due to interference, diffraction and polarization.
2. CO 2: Apply the basic concepts of crystal structures and X-ray diffraction to study the behavior of materials for engineering applications.
3. CO 3: Summarize the fundamental properties of dielectric and magnetic materials for engineering applications.
4. CO 4: Analyze the properties of quantum particles to interpret the energy band formation and classification of solids
5. CO 5: Assess the current flow mechanism to understand the transport phenomenon of charge carriers in semiconductors.

5. Co-PO/PSOMapping

APPLIED PHYSICS	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	3	3												
CO2	3	3												
CO3	3	3												
CO4	3	3												
CO5	3	3												

6. LessonPlan

Lecture No.	Weeks	Introductionto dielectrics	References
1	1	Introductiontodielectrics	T1, R1
2		Basicdefinitionsofdielectrics	T1, R1
3		Relationbetweensusceptibilityanddielectricconstant,	T1, R1
4		IonicandOrientationPolarization	T1, R1
5		Lorentzfield(Internalfield)	T1, R1

6	2	Frequency dependence of polarization and Clausius-mossotti relation	T1, R1
7		Applications of Dielectrics-Ferroelectricity	T1, R1
8		Introduction and basic definitions of magnetic materials	T1, R1
9	3	Origin of permanent magnetic moment	T1, R1

10		Classification of Magnetic materials	T1, R1
11		Hysteresis-soft and hard magnetic materials	T1, R1
12		Weiss theory of ferromagnetism (qualitative)	T1, R1
13	4	Applications	T1, R1
14		Problems	T1, R1

7. Activity Based Learning

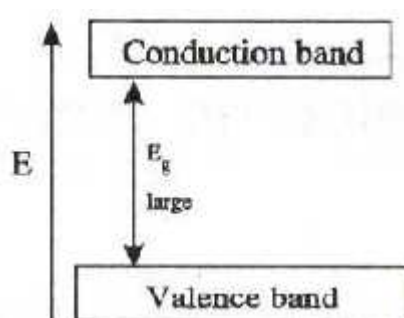
1. Determination of dielectric constant by charging and discharging method.
2. Study the variation of B versus H by magnetizing the magnetic material (B-H curve)

8. Lecture Notes

8.1. Introduction:

By combining two words Dia and electric we get the word "dielectric". The Greek prefix di means "Through". Electric means "electric field".

According to band theory of solids, solids can be classified into conductors, semiconductors and insulators. In insulators, the valence band is full while the conduction band is empty. Further the energy gap between valence band and conduction band is very large, i.e., greater than 3 eV. So electrons cannot jump from the valence band into the conduction band. This means that there are no free electrons available for conduction in the insulators under normal conditions.



For this reason, the electrical conductivity is extremely small and may be nil under ordinary conditions. Though some of the insulators exhibit behavior of electric polarization when they are placed in an electric field, such insulators are known as

dielectrics. Therefore, a dielectric is an electrical insulator that can be polarized by an applied electric field.

In other words, if the main function of non-conducting materials is to provide electrical insulation, then they are called as insulators. On the other hand, if the main function of non-conducting materials is to store electrical charges, then they are called as dielectrics. Dielectrics are widely used in electrical applications.

8.1.1. Dielectrics

Dielectrics are non-conductors of electricity which do not contain free charge carriers.

Examples: air, mica, rubber, ceramics, glass, wood and plastic etc.

Properties

1. They are insulators.
2. They have a very large energy gap (more than 3 eV)
3. All the electrons in the dielectrics are tightly bound to their parent nucleus.
4. As there are no free electrons to carry the current, the electrical conductivity of dielectrics is very low.
5. They have high specific resistance.
6. They have a negative temperature coefficient of resistance.
7. They can be polarized by an electric field.
8. The main function of dielectric materials is to store electric energy.

8.1.2. Types of Dielectrics

Each atom/molecule of a dielectric is neutral. Depending on the atomic/molecular structure, dielectrics are classified into two types. They are

- i) Non-polar Molecules
- ii) Polar Molecules

Non-polar Molecules

If the centers of gravity of positive and negative charges in the molecules coincide; so that no electric dipoles are formed, the molecules of the dielectrics are said to be non-polar molecules. The examples of non-polar molecules are H_2 , N_2 , O_2 , CH_4 and CO_2 etc.

Polar Molecules

If the centers of gravity of positive and negative charges in the molecules do not coincide, so that electric dipoles are formed, the molecules of the dielectrics are said to be polar molecules. The examples of polar molecules are H_2O , HCl , NH_3 , CH_3Cl and CO etc.

Though each molecule of a polar dielectric has its own dipole moment; but the molecular dipoles are randomly oriented in all directions, so that the net dipole moment as a whole is zero.

8.1.3 Basic definitions

1. Electric dipole

Two equal and opposite electric charges are separated by a small distance is called an electric dipole.

Consider two opposite charges of magnitudes $+q$ and $-q$ are separated by a small distance d as shown in Fig.2.1. These two electric charges constitute an electric dipole. Some of the examples of the electric dipoles are HCl , CO_2 and Water etc.

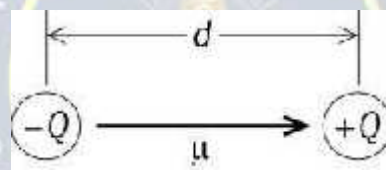


Fig.2.2. Electric dipole

2. Electric dipole moment (μ)

If two opposite charges are separated by a certain distance, a dipole moment μ arises.

Mathematically, it is the product of magnitude of charge & distance of separation between the charges.

$$\mu = q \cdot d$$

→ (2.1)

It is a vector quantity pointing from a negative charge towards a positive charge. The S.I. unit of Dipole moment is coulomb-meter (C-meter) or Debye.

$$1 \text{ debye} = 3.3 \times 10^{-30} \text{ C-m}$$

8.2. Polarizationindielectrics

Theprocessofproducingelectricdipolesbyanelectricfieldiscalledpolarizationindielectrics.

When an electric field is applied to the dielectrics, the field exerts a force on eachpositivechargeinitssowndirection,asresulttthepositivechargesaredisplacedinthedirection of field while negative charges are displaced in the opposite direction.Consequently, the displacement of these charges produces electric dipoles throughoutthedielectricmaterial.Thisprocessisknownaspolarizationindielectrics.

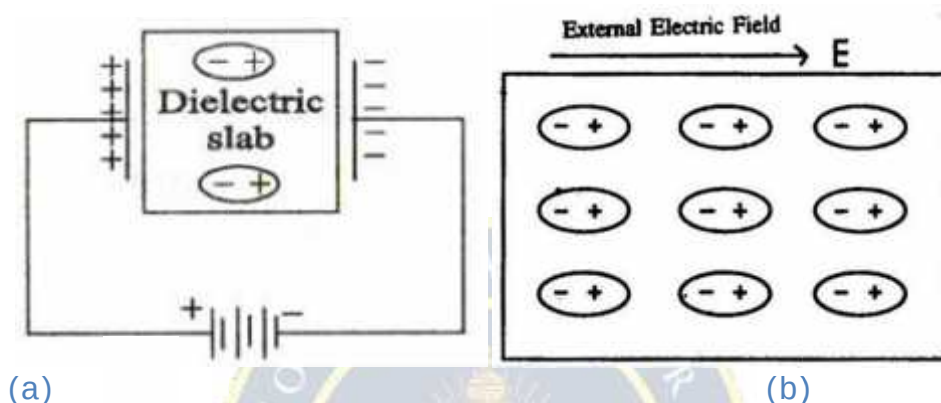


Fig.2.3.Polarizationindielectrics.

8.2.1. Polarizability(α)

If the strength of the electric field E is increases, the strength of the induced dipolemomentalsoincreases.Theinduceddipolemomentisproportionaltotheintensityofelectricfield.

$$\vec{\mu} = \alpha E \rightarrow (2.2)$$

Where α isthe proportionalityconstant,calledPolarizability.

8.2.3.Polarisationvector($\vec{P}$)

If μ is the average dipole moment per molecule and N is the number of moleculesperunitvolume,the polarizationvector($\vec{P}$)isdefinedasdipolemomentperunitvolume ofthedielectricmaterial.

$$\vec{P} = N\vec{\mu} \rightarrow (2.3)$$

Units:Coulomb/m²

8.2.3 Electric susceptibility(χ_e)

Theelectricsusceptibility χ_e ofadielectricmaterialisameasureofhoweasilyitpolarizes inresponsetoanelectricfield.

When a dielectric material is placed in an electric field E , then polarization takes place. The polarization vector $\vec{P}$ is proportional to the electric field E .

$$\vec{P} \propto E$$

$$\vec{P} = \chi E \rightarrow (2.4)$$

Where χ is called electric susceptibility. Units: No units

8.2.4 Permittivity

It is defined as the ability of the material to permit the passage of electric field through it. The permittivity of any dielectric material can be represented as $\epsilon = \epsilon_r \epsilon_0$.

Where ϵ_r is called the relative permittivity or dielectric constant of the dielectric material and ϵ_0 is permittivity of free space and is equal to $8.85 \times 10^{-12} \text{ F/m}$.

It is a dimensionless quantity.

8.3. Dielectric constant

It is defined as the ratio of permittivity of medium to permittivity of free space.

$$\epsilon_r = \frac{\epsilon}{\epsilon_0} \rightarrow (2.5)$$

Units: No units

The dielectric constant can also be obtained from electric flux density (D) and an applied electric field (E). The number of electric force lines passing per unit area perpendicular to field is called electric flux density (D). It is proportional to the applied electric field (E).

$$D \propto E$$

$$D = \epsilon E \text{ (in medium)} \rightarrow (2.6)$$

Where ϵ is the proportionality constant, called permittivity of medium.

$$D = \epsilon_0 E \text{ (in free space)} \rightarrow (2.7)$$

8.4 Types of polarization mechanisms:

When a dielectric material is placed in an external electric field, it gets polarized. The four types of polarization which occur in dielectrics are:

1. Electronic polarization,
2. Ionic polarization,
3. Orientation or dipole polarization,

4. Spacechargeorinterfacialpolarization

Let us discuss them one by one:

8.4.1. Electronic polarization

When electric field applied is applied on dielectric material then all the positive nuclei of atoms move in the field direction and the negative electron cloud of atoms move in opposite direction, hence dipoles will be formed. This phenomenon is known as electronic polarization.

Examples: Mono atomic gases exhibit only electronic polarization: Helium (He), Neon (Ne), Argon (Ar), Krypton (Kr), Xenon (Xe) and Radon (Rn).

The induced dipole moment (μ_e) is proportional to the electrical field strength (E). i.e., $\mu_e \propto E$

$$\mu_e = \alpha_e E \rightarrow (2.8)$$

Where α_e is proportionality constant and it is known as electronic polarizability. The electronic polarizability for rare noble gas atoms is given by



$$\alpha_e = A_{\text{He}} \left(\frac{Z}{R} \right)^3$$

8.4.2. Ionic polarization

When an electric field is applied on ionic dielectric material then positive ions move in the field direction where as negative ions move in the opposite direction, hence dipoles will be formed. This phenomenon is known as ionic polarization.

Examples: Ionic solids (NaCl) exhibit ionic polarization.

The induced dipole moment (μ_i) due to ionic polarization is proportional to the electrical field strength (E).

$$\text{i.e., } \mu_i \propto E$$

$$\mu_i = \alpha_i E$$

Where α_i is proportionality constant and it is known as ionic polarizability. The ionic polarizability for an ionic solid is given by



$$\alpha_i = \frac{4\pi}{3} \left(\frac{V}{N} \right) \left(\frac{Z}{R} \right)^3$$

8.4.3 Orientation polarization (Dipolar polarization)

Orientation polarization arises in dielectric materials which possess molecules with permanent dipole moment (i.e., in polar molecules).

Examples: Dipolar polarization or orientation takes place only in polar molecules such as H_2O , HCl , NH_3 , CH_3Cl and CO etc.

The induced dipole moment (μ_0) due to orientation polarization is proportional to the electrical field strength (E).

$$\text{i.e., } \mu_0 \propto E$$

$$\mu_0 = \alpha_0 E$$

Where α_0 is proportionality constant and it is known as orientation polarizability. The orientation polarizability for polar dielectric medium is given by

$$\alpha_0 = \frac{4\pi N}{3} \frac{\mu^2}{kT}$$

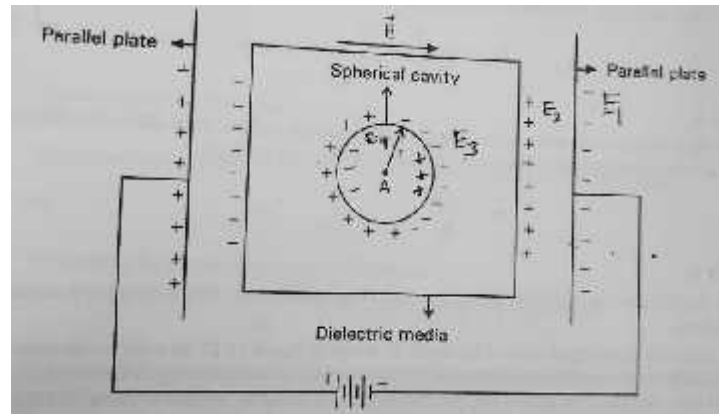
8.5. Local (internal) field or Lorentz relation

Definition

When a dielectric material is subjected to an external electric field, each of the atoms develops a dipole moment and acts as an electric dipole. Hence the resultant field at a given atom will be the sum of applied electric field and the electric field due to surrounding dipoles. This resultant field acting at an atom in a dielectric is called local field (or) internal field E_{int} and is different from the applied external field E_{app} . This was first calculated by Lorentz.

Calculation of local field (or) internal field (E_{int})

To calculate an expression for local electric field on a dielectric molecule or an atom, we consider a dielectric material in the electric field of intensity E , between the capacitor plates so that the material is uniformly polarized, as a result opposite type of charges are induced on the surface of the dielectric near the capacitor plates as shown in Fig. 4.9.

Fig.4.9.Localfield(or)Internalfield E_{int}

Let us assume a small sphere region or cavity of radius 'r' around the atom A inside the dielectric at which the local field is to be calculated. It is also assumed that the radius of the cavity is large compared to the radius of the atom. i.e., there are many atomic dipoles within the sphere.

The electric field acting on the central atom of the sphere is called local field (or) internal field, which is a result due to the sum of the following fields.

$$E_{int} = E_1 + E_2 + E_3 + E_4 \rightarrow (2.30)$$

Where,

E_1 = Field at A due to the charges on the plates (externally applied).

E_2 = Field at A due to the polarized charges induced on the two sides of the dielectric.

E_3 = Field at A due to the polarized charges induced on the surface of the spherical cavity.

E_4 = Field at A due to the atomic dipoles inside the spherical cavity. Now, let us calculate E_1, E_2, E_3, E_4 values one by one as follows.

Field E_1

When a dielectric medium is polarized due to an electric field E , the displacement vector D is given by

$$D = \epsilon_0 E + P \rightarrow (2.31)$$

From the field theory, the field at A due to the charges on the plates is given by

$$D = \epsilon_0 E_1 \rightarrow (2.32)$$

$\therefore$ Equating equation (2.31) and (2.32), we get

$$\epsilon_0 E_1 = \epsilon_0 E + P$$

Dividing the above equation by ϵ_0 , we get

$$E_1 = E + \frac{P}{\epsilon_0} \rightarrow (2.33)$$

Field E_2

E_2 is the field at A due to the polarized charges induced on the two sides of dielectric. This field acts in a direction opposite to the external field. From the field theory, we have

$$E_2 = - \frac{P}{\epsilon_0} \rightarrow (2.34)$$

Field E_3

E_3 is the field at A due to the polarized charges induced on the surface of the spherical cavity. This field value was calculated by Lorentz as given below.

Consider a magnified or enlarged view of the imagined spherical cavity as shown in

Fig.4.10. A small elemental ring is cut with area 'dA' perpendicular to the field direction and is making between θ and $\theta + d\theta$, where θ represents the direction with respect to the applied field direction.

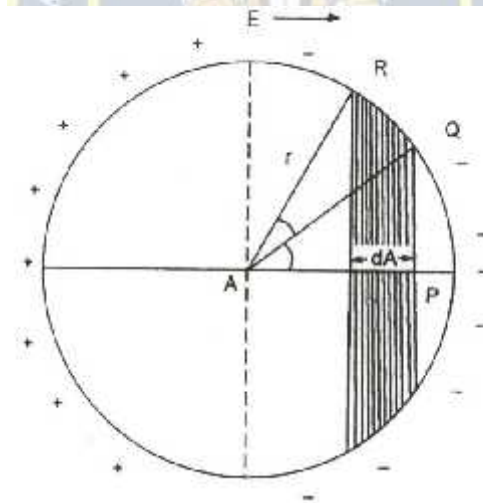


Fig.4.10. Enlarged view of the imagined spherical cavity

From the figure, the area of the small elemental ring along the surface of the sphere is given by

$$dA = 2\pi(PQ)(QR) \rightarrow (2.35)$$

From the triangle APQ, $\sin \theta = \frac{PQ}{r}$ or $PQ = r \sin \theta \rightarrow (2.36)$

and from the sector AQR, $d\theta = \frac{QR}{r}$ or $QR = r d\theta \rightarrow (2.37)$

Substituting equations (2.36) and (2.37) in equation (2.35)

$$dA = 2\pi(r \sin \theta)(r d\theta)$$

$$dA = 2\pi r^2 \sin \theta d\theta \rightarrow (2.38)$$

From the definition of polarization (i.e., $P = \frac{q}{A}$), The charge q on the surface A is equal to the product of the normal component of the polarization and the surface area.

$$\text{i.e., } dq = P_N \cdot dA \rightarrow (2.39)$$

The polarization P is parallel to E . Its component normal to dA is $P_N = P \cos \theta$

$$\cos \theta \cdot dq = P \cos \theta dA \rightarrow (2.40)$$

Substituting dA value in equation (2.40)

$$dq = P \cos \theta \cdot 2\pi r^2 \sin \theta d\theta \rightarrow (2.41)$$

The field due to the charge q is denoted by dE_3 at 'A' is given by (Coulomb's law)

$$dE_3 = \frac{dq \cos \theta}{4\pi \epsilon_0 r^2} \rightarrow (2.42)$$

Substituting equation (2.41) in equation (2.42)

$$dE_3 = \frac{(P \cos \theta \cdot 2\pi r^2 \sin \theta d\theta) \cos \theta}{4\pi \epsilon_0 r^2} = \frac{P \cos^2 \theta \sin \theta d\theta}{2\epsilon_0} \rightarrow (2.43)$$

Therefore, the field due to the surface charge on the cavity is obtained by integrating over the whole surface of the sphere.

$$\int_0^\pi dE_3 = \frac{P}{2\epsilon_0} \int_0^\pi \cos^2 \theta \sin \theta d\theta \rightarrow (2.44)$$

$$E_3 = \frac{P}{2\epsilon_0} \int_0^\pi \cos^2 \theta \sin \theta d\theta \rightarrow (2.45)$$

$$= \frac{P}{2\epsilon_0} \left[-\frac{\cos^3 \theta}{3} \right]_0^\pi = \frac{P}{2\epsilon_0} \left(-\frac{\cos^3 \pi}{3} + \frac{\cos^3 0}{3} \right) = \frac{P}{2\epsilon_0} \left(\frac{1}{3} + \frac{1}{3} \right) = \frac{P}{\epsilon_0}$$

$$\therefore E_3 = \frac{P}{\epsilon_0} \rightarrow (2.46)$$

Field E_4

E_4 is the field due to the atomic dipoles inside the spherical cavity which depends on the crystal structure. This field E_4 is zero for spherically symmetric system (cubic structure) because the dipoles will cancel with each other.

$$\therefore E_4 = 0 \rightarrow (2.47)$$

Hence, substituting all the four field values from the equations (2.33), (2.34), (2.46) and (2.47) in equation (2.30)

$$E_{int} = E + \frac{P}{\epsilon_0} - \frac{P}{\epsilon_0} + \frac{P}{3\epsilon_0} \rightarrow (2.48)$$

$$E_{int} = E + \frac{P}{3\epsilon_0} \rightarrow (2.49)$$

This equation is known as Local field or internal field. Thus, it is observed that the local field is greater than the electric field applied by an additional factor

$$\frac{P}{3\epsilon_0}$$

8.6. Clausius-Mosotti Equation

The relation between dielectric constant ϵ_r (macroscopic quantity) and polarizability α (microscopic quantity) of atoms in a dielectric is known as Clausius-Mosotti Equation.

$$\frac{\epsilon_r - 1}{3\epsilon_0} = \frac{N\alpha}{\epsilon_0} + 2$$

This relation is known as Clausius-Mosotti Equation.

Proof

When a dielectric material is placed in an external electric field E , it gets polarized, i.e., dipole moments are induced. If the strength of the electric field E is increased, the strength of the induced dipole moment also increases. The induced dipole moment is proportional to the intensity of electric field.

$$\vec{\mu} = \alpha \vec{E}_{int}$$

$$\vec{\mu} = \alpha \vec{E}_{int} \rightarrow (2.50)$$

Where α is the proportionality constant, called Polarizability and E_{int} is the local field.

If μ is the average dipole moment per molecule and N is the number of molecules per unit volume, the polarization vector ($\vec{P}$) is defined as dipole moment per unit volume of the dielectric material

$$P = N\mu \rightarrow$$

(2.51) Substituting equation (2.50) in equation (2.51), we have

$$P = N\alpha E_{\text{int}} \rightarrow (2.52)$$

We know that

$$E_i = E + \frac{P}{3\epsilon_0}$$

$$P = N\alpha \left(E + \frac{P}{3\epsilon_0} \right)$$

$$P = N\alpha E + N\alpha \frac{P}{3\epsilon_0}$$

$$P - N\alpha \frac{P}{3\epsilon_0} = N\alpha E$$

$$\left(1 - \frac{N\alpha}{3\epsilon_0} \right) P = N\alpha E$$

$$P = \frac{N\alpha E}{1 - \frac{N\alpha}{3\epsilon_0}} \rightarrow (2.53)$$

We also know that

$$P = \epsilon_0 E (\epsilon_r - 1) \rightarrow (2.54) \text{Equation (2.53) and (2.54)}$$

ating the equations (2.53) and (2.54)

$$\epsilon_0 E (\epsilon_r - 1) = \frac{N\alpha E}{1 - \frac{N\alpha}{3\epsilon_0}}$$

$$1 - \frac{N\alpha}{3\epsilon_0} = \frac{N\alpha E}{\epsilon_0 E (\epsilon_r - 1)}$$

$$1 = \frac{N\alpha E}{\epsilon_0 E (\epsilon_r - 1)} + \frac{N\alpha}{3\epsilon_0}$$

$$1 = \frac{N\alpha}{\epsilon_0} \left[\frac{1}{(\epsilon_r - 1)} + 1 \right]$$

$$\frac{N\alpha}{3\epsilon_0} = \frac{1}{\frac{\epsilon_r - 1}{1 + \epsilon_r - 1} + 1}$$

$$\frac{N\alpha}{3\epsilon_0} = \frac{\epsilon_r - 1}{3 + \epsilon_r - 1}$$

$$\frac{N\alpha}{3\epsilon_0} = \frac{\epsilon_r - 1}{\epsilon_r + 2} \rightarrow (2.55)$$

The above relation is known as Clausius-Mosotti equation.

8.7. Ferroelectricity

In general, when the dielectric materials are placed in an external electric field E , they get polarized. But some dielectric materials exhibit spontaneous polarization even in the absence of an external electric field. This phenomenon is known as ferroelectricity and these dielectric materials are known as ferroelectrics.

The ferroelectric effect was first observed by Valasek in 1921, in the Rochel salt. This has molecular formula $\text{KNaC}_4\text{H}_4\text{O}_6 \cdot 4\text{H}_2\text{O}$.

Examples:

1. Rochel salt ($\text{NaKC}_4\text{H}_4\text{O}_6 \cdot 4\text{H}_2\text{O}$)
2. Potassium dihydrogen phosphate or KDP (KH_2PO_4)
3. Ammonium dihydrogen phosphate or ADP ($\text{NH}_4\text{H}_2\text{PO}_4$)
4. Potassium niobate (KNbO_3)
5. Lithium niobate (LiNbO_3)
6. Barium titanate (BaTiO_3)
7. Lithium titanate (LiTiO_3) etc.

Properties

1. They are anisotropic crystals.
2. They exhibit spontaneous polarization.
3. As the temperature increases the spontaneous polarization decreases and at a particular temperature, the spontaneous polarization vanishes. This temperature is known as Curie temperature (T_c). The variation of spontaneous polarization with temperature as shown in Fig.
4. Below Curie temperature, the dielectric constant is function of electric field. Above Curie temperature dielectric constant varies with temperature. The variation of dielectric constant with temperature as shown in Fig.

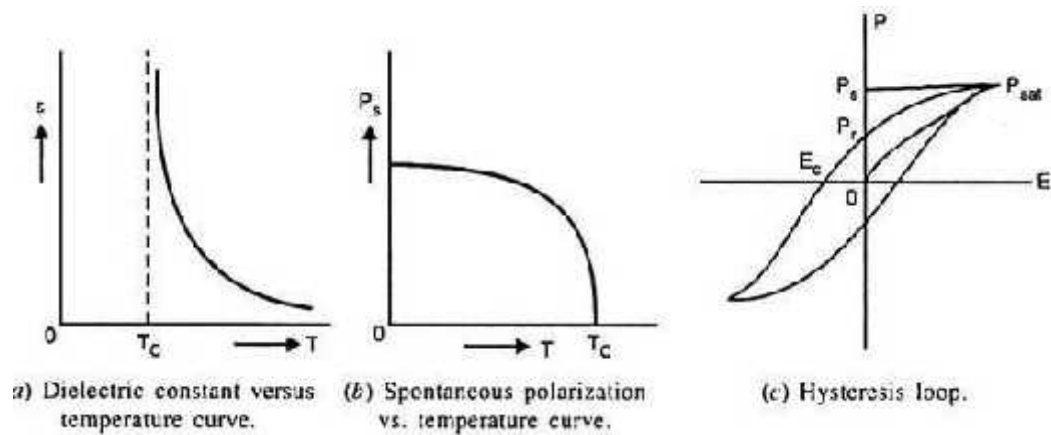
According to Curie–Weiss law,

$$\epsilon_r = \frac{C}{T - T_c}$$

Where C is called Curie constant and T_c is called Curie temperature.

5. Ferroelectric materials exhibit hysteresis, similar to that of ferromagnetic materials.

Hysteresis: When an electric field is applied on a ferroelectric material then polarization takes place. This polarization always lags behind the applied electric field. This phenomenon is known as hysteresis of ferroelectric material.



4.11. Properties of ferroelectrics

6. Ferroelectric materials exhibit piezoelectricity and pyroelectricity.

When a dielectric material acquires electric polarization due to the external mechanical pressure, it is called piezoelectricity and these materials are known as piezoelectrics.

When a dielectric material acquires electric polarization due to the external thermal energy, it is called pyroelectricity and these materials are known as pyroelectrics.

8.7.1 Applications of Ferroelectric Materials

Ferroelectric materials can be used in various applications; some of the applications are given below.

1. Production of ultrasonics.
2. SONAR, strain gauges, etc.
3. Frequency stabilizers and crystal-controlled oscillators.
4. Magnetic amplifiers and dielectric amplifiers.
5. Magnetic information storage devices.
6. Magnetostrictive transducers.
7. Capacitor microphones.
8. Gas filters.
9. Delay lines.
10. High sensitive infra-red detectors.

8.8. Introduction:

Magnetism

A substance that attracts pieces of iron (or) steel is called "Magnet". This property of a substance is called "magnetism".

Magnetic Poles:

When a bar magnet is dipped in a heap of iron filings and taken out, it is seen that maximum amount of iron filings are seen to cling to the magnet at the two ends of the magnet. Practically no iron filings cling in the middle. The regions of the magnet at which maximum amount of iron filings cling are called poles of the magnet. Poles of magnet are regions near the two ends of a magnet with maximum power of attraction. The strength of the pole is called pole strength denoted by m . The S.I. unit of pole strength is Ampere Meter.

The distance between two magnetic poles is called "magnetic length" ($2l$).

Magnetic Dipole:

- Magnetic dipoles are found to exist in magnetic materials, analogous to electric dipoles.
- Two equal and opposite charges separated by a small distance is called an electric dipole. Similarly a north pole and south pole separated by a small distance $2l$ (magnetic length) constitute a magnetic dipole.
- For example: A bar magnet, a compass needle etc. are the magnetic dipoles. And also a current loop behaves as a magnetic dipole.

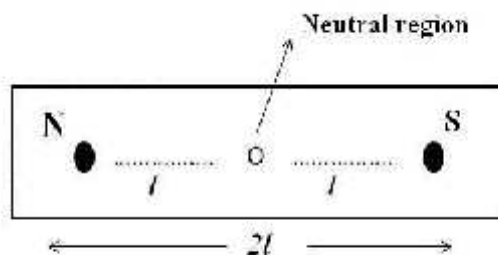


Fig : Magnetic dipole

Magnetic Dipole Moment:

The behavior of magnetic dipole is described by the magnetic dipole moment.

(a) In the case of bar magnet:

It is defined as the product of pole strength (m) and magnetic length ($2l$).



$$\mu_m = m(2l)$$

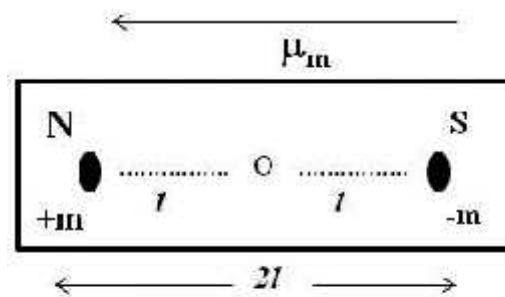


Fig : Magnetic dipole moment

It is a vector quantity. It is directed from South Pole to North Pole. The S.I. Unit of magnetic dipole moment: Ampere-meter² (A-m²).

(b) In the case of current loop:

A current carrying loop behaves as a magnetic dipole.

Consider a current carrying conductor loop of wire as shown fig. The current (I) establishes a magnetic field around the loop. By right hand palm rule, the upper face of the loop acts as a S-pole and the lower face acts as a N-pole.

The magnitude of dipole moment of current loop (μ_m) is

- (i) Directly proportional to current (I) through the loop.
- (ii) Directly proportional to the area of cross-section (A).

$$\mu_m \propto I \cdot A$$

$$\mu_m = K I A$$

$$\mu_m = I A$$

Where K is a proportionality whose value is one

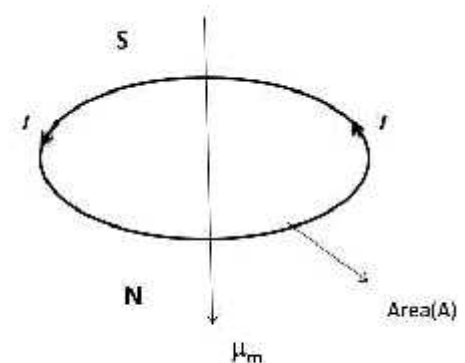


Fig : Current carrying conductor loop

Magnetic Field:

The space surrounding a magnet where magnetic force is experienced is called a magnetic field.

A magnetic field can be represented by drawing lines called "magnetic lines of force". The lines go from North to South on the magnet.

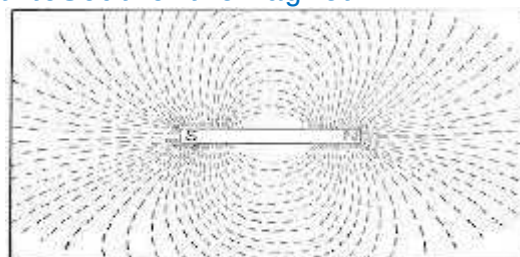


Fig: Magnetic field.

Magnetic Flux:

A group of magnetic lines of force is called "magnetic flux". The symbol for magnetic flux is Φ (phi).

The SI unit of magnetic flux is the Weber (Wb). One Weber is equal to 1×10^8 magnetic field lines.

Magnetic Flux Density:

Magnetic flux density is the amount of magnetic flux per unit area of a section, perpendicular to the direction of flux.

$$\text{Magnetic flux density (B)} = \frac{\text{Magnetic flux (Weber)}}{\text{Area (m}^2\text{)}}$$

$$B = \frac{\Phi}{A} \text{ Tesla}$$

Magnetization:

Magnetization in magnetic field is analogous to polarization of dielectric material in electrostatic field. The process of converting a non-magnetic material into a magnetic material is known as "magnetization".

Intensity of Magnetization (I or M)

When a material medium is placed in a magnetic field, it gets magnetized. To magnetize a material medium

into a magnetic dipole moments.

The magnetic dipole moment per unit volume of the material is called the intensity of magnetization I (or simply magnetization).

$$I = \frac{\text{Magnetic dipole moment } (\mu_m)}{(\text{m}) \text{ Volume (V)}} = \frac{\text{Length of magnet (2l)} \times \text{Pole Strength}}{\text{Length of Magnet (2l)} \times \text{Area of cross-section (A)}} = \frac{\text{Pole Strength (m)}}{\text{Area of Cross Section (A)}}$$

The SI unit of magnetization is ampere/meter.

Magnetic Field Strength (H):

The ability of magnetic field to magnetize a material medium is called its magnetic intensity or field strength. It is denoted by H.

The SI unit of magnetic field strength is ampere/meter.

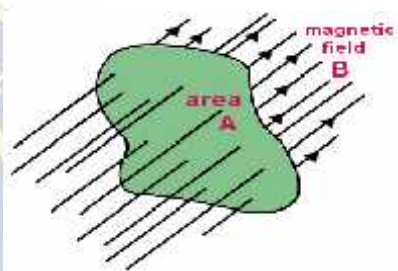
8.9. Magnetic Susceptibility (χ_m):

The word Susceptibility comes from the Latin word "susceptible" meaning the easily affected.

The magnetic susceptibility of a material medium indicates how easily a material medium can be magnetized in the presence of magnetic field..

The intensity of Magnetization is directly related to the applied field strength H.

$$M \propto H$$



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$$M = \chi_m H$$

$$\text{Magnetic Susceptibility}(\chi_m) = \frac{M}{H}$$

Therefore; the magnetic susceptibility of a material is defined as the ratio of intensity of magnetization (I) developed in the material to the applied magnetic field (H).

Magnetic Permeability (μ):

- It is the natural property of material.
- In Latin, perme means through and mea means to pass.
- It is defined as the ability of the material to permit the passage of magnetic lines of force through it.
- The Magnetic induction B is proportional to the applied Magnetic field intensity H. $B \propto H$

$$B = \mu H$$

$$\text{Magnetic Permeability}(\mu) = \frac{B}{H}$$

Where " μ " is the permeability of a

medium. For vacuum, $B = \mu_0 H$

Where μ_0 is the proportionality constant and is also called permeability of the free space and its value is $4\pi \times 10^{-7} \text{ Hm}^{-1}$.

Relative permeability (μ_r):

The ratio of permeability of medium to the permeability of free space is called relative permeability μ_r of the medium

$$\mu_r = \frac{\mu}{\mu_0}$$

$\mu_r = 1$ for vacuum. It has no units.

The Relation between Relative Permeability and Magnetic Susceptibility:

When a magnetic material is magnetized by placing it in a magnetic field, the resultant field inside the material is the sum of the field due to the magnetization of the material and the original magnetizing field. The resultant field is called magnetic induction or magnetic flux density B.

$$B = \mu_0 H + \mu_0 M$$

$$B = \mu_0 (H + M)$$

$$\mu_0 \mu_r H = \mu_0 (1 + \chi_m) H \quad (B = \mu H \text{ and } \mu = \mu_0 \mu_r)$$

$$\text{Where } \chi_m = \frac{M}{H}$$

$$\mu_r = (1 + \chi_m)$$

This is the relation between Relative Permeability and Magnetic Susceptibility.

8.10. Origin of Magnetism:

Magnetism originates from magnetic dipole moment. This magnetic dipole moment arises due to the rotational motion of charged particles.

According to modern view:

- All substances are made of atoms or molecules. An atom which consists of a positively charged nucleus at the centre and negatively charged electrons revolving around the nucleus in different orbits. This motion of electrons is called orbital motion as shown in fig. The orbiting electrons constitute tiny current loops. These loops behave as the magnetic dipoles.
- The orbital motion of electrons around the nucleus gives rise to the orbital magnetic dipole moment (μ_{orbit}).
- The electrons also rotate around their own axes. This motion of electrons is called spin motion as shown in fig. The spinning motion of electrons around their axes gives rise to the spin magnetic dipole moment (μ_{spin}).
- The motion of the protons and neutrons within the nucleus also contributes to the total magnetic moment (μ_{nucleus}). But the magnitude of the nuclear magnetic moment is (about 10^{-3} times) very small compared with the magnetic moment of electron and is usually neglected.

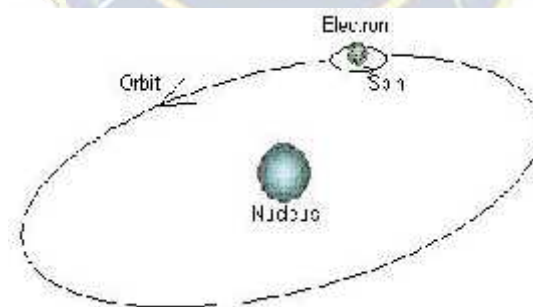


Fig: Motion of electron

Orbital magnetic dipole moment of electron (μ_{orbit}):

The magnetic dipole moment arises due to the orbital motion of electrons around the nucleus is called orbital magnetic dipole moment (μ_{orbit}).

Proof:

$$\mu_{\text{Orbit}, z} = -\mu_B \cdot m_l$$

Let us consider an electron of mass 'm' and charge 'e' revolving around the nucleus in a circular orbit of radius 'r' with linear velocity 'v' as shown in fig.

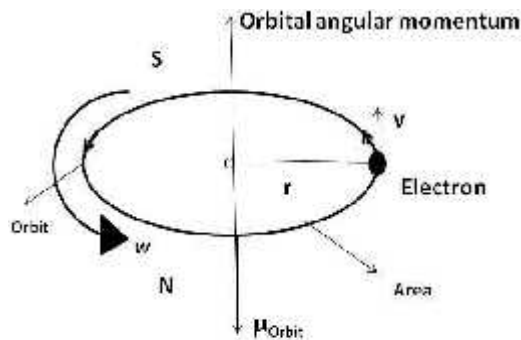


Fig : Orbital motion of electron

The revolving electron in circular orbit establishes a current given by

$$I = \frac{\text{Charge of electron}}{\text{time period}} = \frac{-e}{T} \quad (1)$$

Where 'T' is the time taken by the electron to make one revolution around the nucleus

$$\text{i.e., } T = \frac{2\pi}{\omega} \quad (2)$$

Where 'ω' is the angular frequency of the electron

But relation between linear velocity 'v' and angular velocity can be written as

$$v = r\omega \quad \text{and } \omega = \frac{v}{r} \quad (3)$$

Substituting the equation (3) in (2),

$$T = \frac{2\pi r}{v} \quad (4)$$

Further, substituting the equation (4) in (1),

$$I = \frac{-ve}{2\pi r} \quad (5)$$

The current 'I' establishes a magnetic field around the circular orbit, so that the upper surface acts as South Pole and the lower surface acts as North Pole.

$$\text{The Area of the orbit is } A = \pi r^2 \quad (6)$$

Then the corresponding magnetic dipole moment is given by

$$\mu_{\text{Orbit}} = IA$$

$$= \frac{-ve \times \pi r^2}{2\pi r} = \frac{-e v r}{2} \quad (7)$$

Dividing and multiplying the equation (7) by the mass 'm' of electron.

$$\mu_{\text{Orbit}} = \frac{-e v r}{2} \times \frac{m}{m} = \frac{-e(mvr)}{2m}$$

$$= \frac{-e(L)}{2m} \quad (\text{But } L = mvr)$$

$$\mu_{\text{Orbit}} = \frac{-e}{2m} L (\text{Orbital angular momentum})$$

The –ve sign indicates that the orbital angular momentum and orbital magnetic dipole moment

are in opposite directions.

An electron has an intrinsic orbital angular momentum (L) itself cannot be measured. However, its component along any axis can be measured. Let us assume that the component of orbital angular momentum (L) is measured along the z-

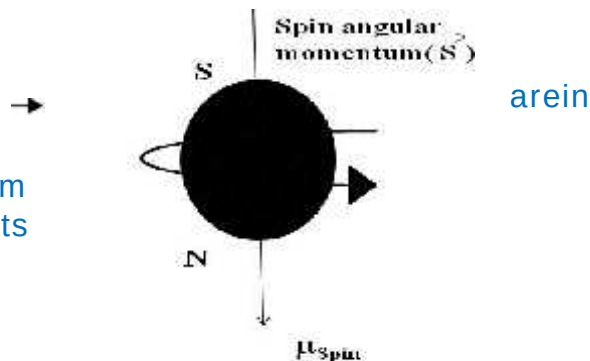


Fig: Spin motion of electron

of a coordinate system. Then the measured component L_z can have only the values given by

$$L_z = m_l \frac{h}{2\pi}$$

Where m_l is called orbital magnetic quantum number $= 0, \pm 1, \pm 2, \pm 3 \dots$

The orbital magnetic dipole moment (μ_{Orbit}) of electron itself also cannot be measured. Only its component along any axis can be measured.

Let us assume that the component of orbital magnetic dipole moment (μ_{Orbit}) of electron is measured along the z-axis of a coordinate system. Then the measured component $\mu_{\text{Orbit},z}$ can have only the two values given by

$$\begin{aligned} \mu_{\text{Orbit},z} &= \frac{-e}{2m} L_z \\ \mu_{\text{Orbit},z} &= -\left(\frac{e}{2m}\right) m_l \frac{h}{2\pi} \\ \mu_{\text{Orbit},z} &= -\left(\frac{eh}{4\pi m}\right) m_l \end{aligned}$$

$$\mu_{\text{Orbit},z} = -\mu_B \cdot m_l$$

Where $\mu_B = \frac{eh}{4\pi m}$ is known as Bohr magneton and its value is $9.27 \times 10^{-24} \text{ A} \cdot \text{m}^2$.

Spin magnetic dipole moment of electron (μ_{spin}):

The magnetic dipole moment arises due to its spin motion is called spin magnetic moment (μ_{spin}) and is given by

$$\mu_{\text{Spin}} = -g \left(\frac{e}{2m}\right) S (\text{Spin angular momentum})$$

The –

ve sign indicates that the spin angular momentum and spin magnetic dipole moment are in opposite directions.

Where g is called Landé's g factor or Spectroscopic splitting factor

$$g = \frac{1+J(J+1)+S(S+1)-L(L+1)}{J(J+1)}$$

$g_l = 1$ for orbital motion
 $g_s = 2$ for spin motion



An electron has an intrinsic spin angular momentum (S) itself cannot be measured. However, its component along any axis can be measured.

Let us assume that the component of spin angular momentum (S) is measured along the z-axis of a coordinate system. Then the measured component S_z can have only the values given by

$$S_z = m_s \frac{h}{2\pi}$$

Where m_s is called magnetic spin quantum number = $\pm \frac{1}{2}$

$m_s = +\frac{1}{2}$ for spin up and

$m_s = -\frac{1}{2}$ for spin down

The spin magnetic dipole moment (μ_{spin}) of electron itself also cannot itself be measured. Only; its component along any axis can be measured. Let us assume that the component of spin magnetic dipole moment (μ_{spin}) of electron is measured along the z-axis of a coordinate system. Then the measured component $\mu_{\text{spin}, z}$ can have only the two values given by

$$\mu_{\text{spin}, z} = -g(e) S_z$$

$$\mu_{\text{spin}, z} = -g(e) m_s \frac{h}{2\pi}$$

$$\mu_{\text{spin}, z} = -2 \left(\frac{eh}{4\pi m} \right) m_s$$

$$\mu_{\text{spin}, z} = -2(\mu_B) m_s$$

Where $\mu_B = \frac{eh}{4\pi m}$ is known as Bohr magneton and its value is $9.27 \times 10^{-24} \text{ A} \cdot \text{m}^2$.

Nuclear spin magnetic dipole moment (μ_{Nuclear}):

The atomic nucleus contains protons and neutrons. They have intrinsic spin.

The spin motion of the protons and neutrons within the nucleus also contributes to the total spin magnetic dipole moment and is given by

$$\mu_{\text{nuclear, spin}} = \frac{eh}{4\pi M_N} = 5.525 \times 10^{-27} \text{ A} \cdot \text{m}^2$$

Where M_N is the Mass of the proton

But the magnitude of the nuclear magnetic dipole moment is (about 10^{-3} times) very small compared with the magnetic dipole moment of electron and is usually neglected.

Therefore, the magnetism mainly arises due to the orbital and spin magnetic dipole moment of electron.

8.10. Classification magnetic materials:

Magnetic materials are classified based on presence or absence of the permanent magnetic dipoles in a material. They are

1. Diamagnetic material
2. Paramagnetic material
3. Ferromagnetic material
4. Antiferromagnetic material and
5. Ferrimagnetic material

Diamagnetic materials:

Those materials which when placed in a magnetic field are weakly or feebly magnetized in a direction opposite that of the applied magnetic field are called diamagnetic materials.

Examples:

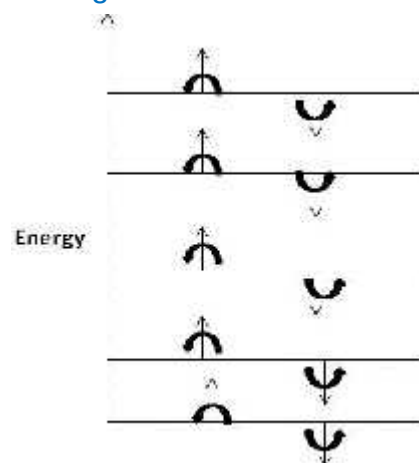
Bismuth, Copper, Zinc, Gold, Water, etc

Cause of diamagnetism:

In diamagnetic materials, there exist paired electrons, so the spins in two opposite directions are equal and hence magnetic dipole moments cancel with each other, i.e., the resultant magnetic dipole moment is equal to zero. Therefore, most of these materials do not have magnetism in the absence of a magnetic field.

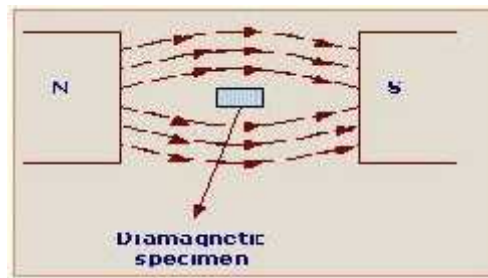
Effect of external magnetic field:

- a) In the absence of an external magnetic field, the atoms/molecule/ion of the diamagnetic substance have no net magnetic dipole moment. Hence, the material does not exhibit diamagnetism.
- b) When a diamagnetic material is placed in an external magnetic field, currents are induced in the current loops of atom/molecule/ion according to Faraday's law of electromagnetic induction. According to Lenz's law, these currents give rise to a magnetic field which opposes the applied magnetic field. Hence, the induced magnetic moments of atoms/molecule/ions are opposite to the applied magnetic field.

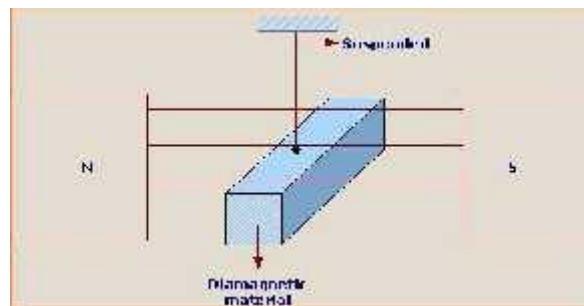


Properties:

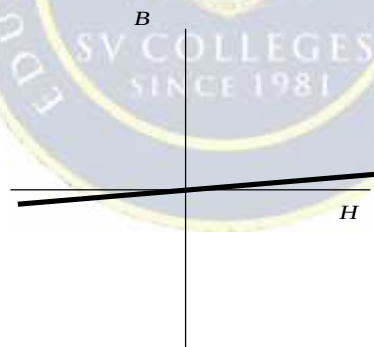
- They do not possess permanent magnetic dipole moment.
- When a diamagnetic material is placed in a magnetic field, it is feebly magnetized in a direction opposite to that of the applied magnetic field.
- When a diamagnetic material is placed in a magnetic field, the magnetic lines of force prefer to pass through the surrounding air rather than through the diamagnetic material.



- The magnetic flux density inside is small than that in the free space. Hence the relative permeability $\mu_r < 1$.
- The magnetic susceptibility (χ_m) is negative and small.
- The magnetic susceptibility (χ_m) is independent of temperature.
- When a rod of diamagnetic material is suspended freely in a uniform magnetic field, the rod comes to rest with its axis perpendicular to the direction of the applied field.



- The B-H curve of diamagnetic material as shown in fig.



Paramagnetic materials:

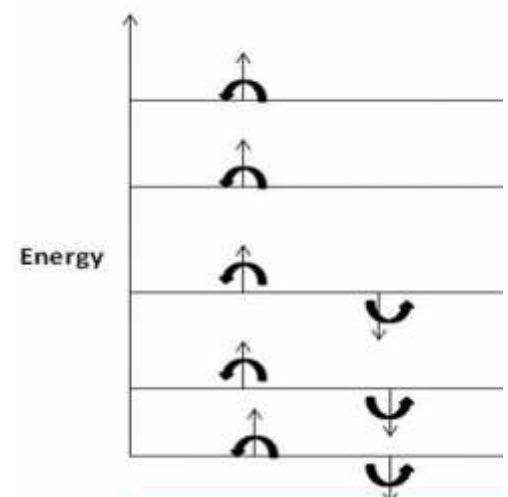
Those materials which when placed in a magnetic field are weakly or feebly magnetized in the direction of the applied magnetic field are called Paramagnetic field materials.

Examples:

Aluminum, platinum, copper sulphate (CuSO_4), manganese, chromium etc.

Cause of paramagnetism:

In the case of paramagnetic materials, the spins in two opposite directions will not be equal. There exists some



unpaired electrons which gives rise to spin magnetic dipole moment.

Hence the resultant magnetic dipole moment will not be equal to zero. i.e., they possess permanent magnetic dipole moment.

Effect of external magnetic field:

- c) In the absence of external magnetic field, the dipoles of the paramagnetic material are randomly oriented and, therefore, the net magnetic dipole moment of the material is zero. Hence, the material does not exhibit paramagnetism.
- d) When a paramagnetic material is placed in an external magnetic field, the magnetic dipoles are partially aligned in the direction of the applied magnetic field. Therefore, the material is weakly or feebly magnetized in the direction of the applied magnetic field.

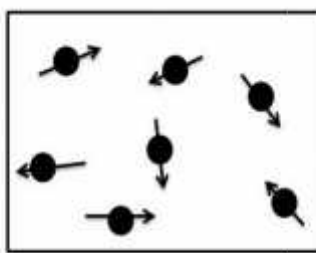


Fig: In the absence of external magnetic field ($H=0$)

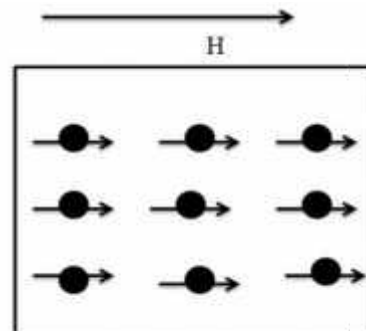
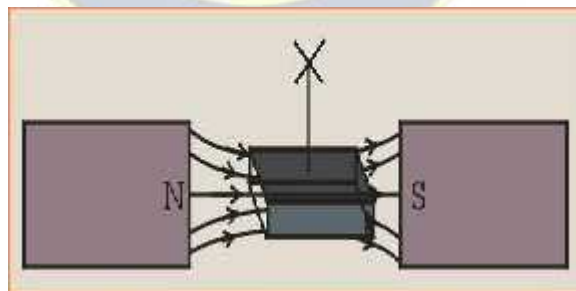


Fig: In the presence of external magnetic field

Properties:

- They possess permanent magnetic dipole moment.
- When a paramagnetic material is placed in a magnetic field, it is feebly or weakly magnetized in the direction of applied magnetic field.
- When a paramagnetic material is placed in a magnetic field, the magnetic lines of force prefer to pass through the paramagnetic material rather than air.

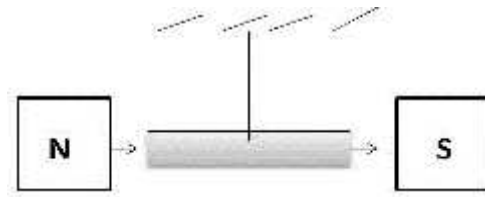


- The magnetic flux density inside is greater than that in the free space. Hence the relative permeability $\mu_r > 1$.
- The magnetic susceptibility (χ_m) is positive and small.
- The magnetic susceptibility (χ_m) is inversely proportional to the temperature.

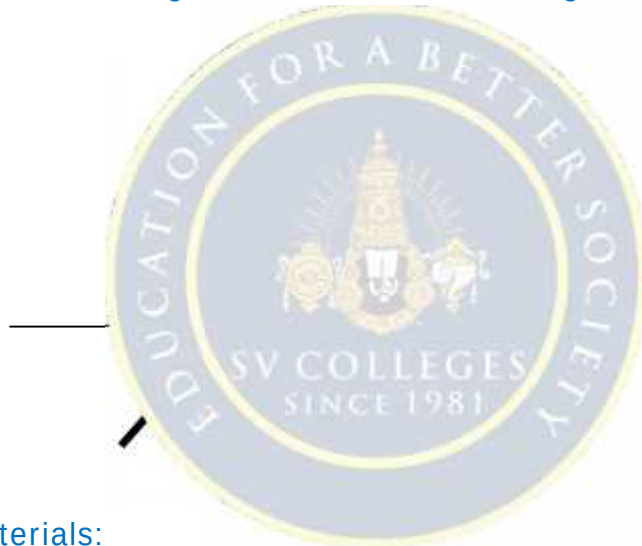
$$\chi_m = \frac{C}{T - \theta_C} \text{ (Curie-Weiss law)}$$

Where $C \rightarrow$ Curie constant
 $T \rightarrow$ Absolute temperature and
 $\theta_c \rightarrow$ Curie temperature

- When the temperature is less than the Curie temperature, paramagnetic materials become diamagnetic material.
- When a rod of paramagnetic material is suspended freely in a uniform magnetic field, the rod comes to rest with its axis parallel to the applied field.



- The B-H curve of Paramagnetic material as shown in fig.



Ferromagnetic materials:

Those materials which when placed in a magnetic field are strongly magnetized in the direction of the applied magnetic field are called Ferromagnetic materials.

Examples:

Iron, Steel, Nickel, Cobalt, etc

Cause of Ferromagnetism:

- In a Ferromagnetic material, the number of unpaired electrons is more and most of the magnetic dipole moments align parallel to each other even in the absence of a magnetic field. Hence they possess permanent magnetic dipole moment even in the absence of a magnetic field.
- In Ferromagnetic materials, atoms are grouped into regions called domains, instead of acting independently like paramagnetic materials. The region of space over which the magnetic dipole moments are aligned is called a domain. A typical domain contains 10^{17} to 10^{21} atoms and occupies a volume of 10^{-12} to 10^{-8} m^3 .

Effect of external magnetic field:

a) In the absence of external magnetic field, the domains of a ferromagnetic material are randomly oriented. In other words, within the domain, all magnetic dipole moments are aligned, but the direction of alignment varies from domain to domain. The result is that there is no net magnetic dipole moment. Therefore, a ferromagnetic material does not exhibit magnetism in the normal state.

b) When a ferromagnetic material is placed in an external magnetic field, a net magnetic dipole moment develops. This can occur in two ways:

- i) By the movement of domain walls
- ii) By the rotation of domain walls.

i) By the movement of domain walls:

- The movement of domain walls takes place in weak magnetic fields.
- Due to a weak magnetic field applied to the material, the magnetic dipole moments increase and hence the boundary of domains is displaced, so that the volume of the domains changes as shown in fig.

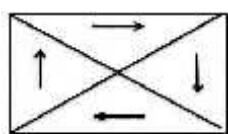


Fig: Without field

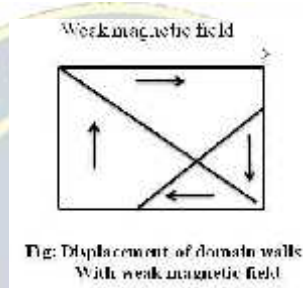


Fig: Displacement of domain walls With weak magnetic field

ii) By the rotation of domain walls

- The rotation of domain wall takes place in strong magnetic fields.
- Due to a strong magnetic field applied to the material, the magnetic dipole moments increase enormously and hence the domains rotate, so that the magnetic dipole moments are aligned in the direction of applied magnetic field as shown in fig.

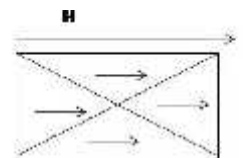
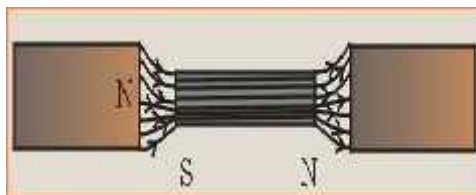


Fig: Rotation of domain walls in strong magnetic field

Properties:

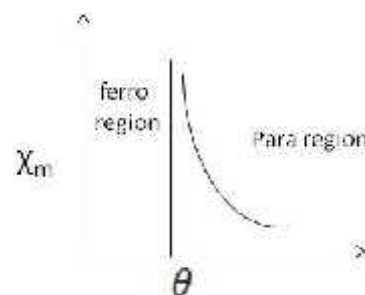
- They possess permanent magnetic dipole moment.
- When a ferromagnetic material is placed in a magnetic field, it is strongly magnetized in the direction of applied magnetic field.
- When a ferromagnetic material is placed in a magnetic field, the magnetic lines of force tend to crowd into the ferromagnetic material.



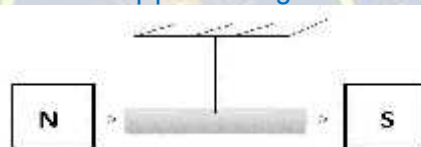
- The magnetic flux density inside is very greater than that in the free space. Hence the relative permeability $\mu_r > 1$.
- The magnetic susceptibility (χ_m) is positive and very high.
- The magnetic susceptibility (χ_m) is inversely proportional to the temperature.

$$\chi_m = \frac{C}{T - \theta_c} \text{ (Curie-Weiss law)}$$

Where $C \rightarrow$ Curie constant
 $T \rightarrow$ Absolute temperature and
 $\theta_c \rightarrow$ Curie temperature



- When the temperature is greater than the Curie temperature, ferromagnetic materials become paramagnetic material.
- When a rod of ferromagnetic material is suspended freely in a uniform magnetic field, it quickly aligns itself in the direction of the applied magnetic field.



Classification of Ferromagnetic materials:

Depending upon the spin orientation of the electrons, ferromagnetic materials are classified into two types, they are 1. Antiferromagnetic materials

2. Ferrimagnetic materials

Antiferromagnetic materials:

The materials which consist of antiparallel spin magnetic dipole moment with same magnitudes are known as antiferromagnetic materials.

Examples:

Ferrous oxide

(FeO), Manganese oxide

(MnO₂), Manganese sulphide

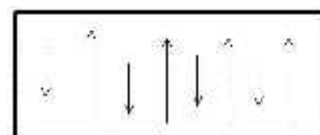
(MnS), Chromium Oxide

(Cr₂O₃), Ferrous Chloride (Fe

Cl₂) etc

Properties:

- In this material, spin magnetic dipole moments are aligned in an antiparallel manner.
- The magnetic susceptibility is very small and positive



- The magnetic susceptibility is inversely proportional to temperature. The variation of susceptibility with temperature is shown in fig.

$$\chi_m = \frac{C}{T + \theta_c}$$

Where $C \rightarrow$ Curie constant
 $T \rightarrow$ Absolute temperature and

$\theta_c \rightarrow$ Curie temperature

χ_m increases gradually with temperature and attains a maximum value at Neel temperature (T_N) and then decreases with increase in temperature.

Ferrimagnetic materials:

The materials which consist of antiparallel magnetic dipole moments of different magnitudes are known as ferrimagnetic materials.

Examples:

Ferrites-general formula: $Me^{+2}Fe_2O_4$

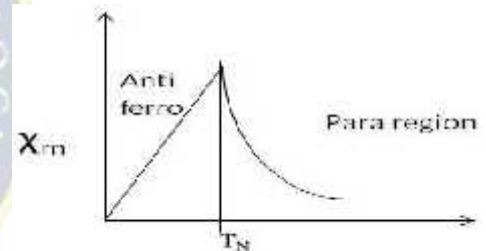
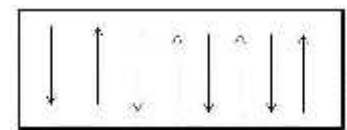
Where Me^{+2} = divalent metal ions (Zn, Cu, Ni).

Properties:

- In this material spin magnetic dipole moments of different magnitudes are aligned in an antiparallel manner.
- The magnetic susceptibility is very high and positive.
- The magnetic susceptibility is inversely proportional to temperature. The variation of susceptibility with temperature is shown in fig.

$$\chi_m = \frac{C}{T + \theta_c}$$

Where $C \rightarrow$ Curie constant
 $T \rightarrow$ Absolute temperature and
 $\theta_c \rightarrow$ Curie temperature.



HYSTERESIS:-

When a ferromagnetic substance (e.g. iron) is subjected to a cycle of magnetization, it is found that flux density B in the material lags behind the applied magnetizing force H . This phenomenon is known as hysteresis.

The term hysteresis is derived from the Greek word *hysterein* meaning to lag behind.

Hysteresis loop:

- If a piece of ferromagnetic material is subjected to one cycle of magnetization, the resultant B - H curve is a closed loop "abcdefa" is called hysteresis loop.
- In fig, O represents unmagnetized position of ferromagnetic material.
 - As H is increased, B increases along oa and reaches its saturation value B_{max} at a . At this stage, all the domains are aligned.
 - If now H is gradually reduced, then it is found that curve follows the path ab instead of ao . At point b , $H=0$ but flux density in the material has a finite value $+B_r (= ob)$ called residual flux density. It is also called remanence or retentivity. Note that B lags behind H . This effect is called hysteresis.

- iv. In order to reduce flux density in the material to zero, it is necessary to apply H in the reverse direction. When H is gradually increased in the reverse direction, the curve follows the path bc . At point c , $B = 0$ and $H = -H_c$. The value of H needed to wipe out residual magnetism is called coercive force (H_c).
- v. Now H is further increased in the reverse direction until point d is reached where the sample is saturated in the reverse direction ($-B_{max}$). If H is now reduced to zero, point e is reached and the sample again retains magnetic flux density ($-B_r$). The remaining part of the loop is obtained by increasing current to produce H in the original direction. The curve "a b c d e f a" is called hysteresis loop. Thus hysteresis loop results because the domains do not become completely unaligned when H is made zero. The area enclosed by the hysteresis loop represents loss in energy. This energy appears in the material as heat.

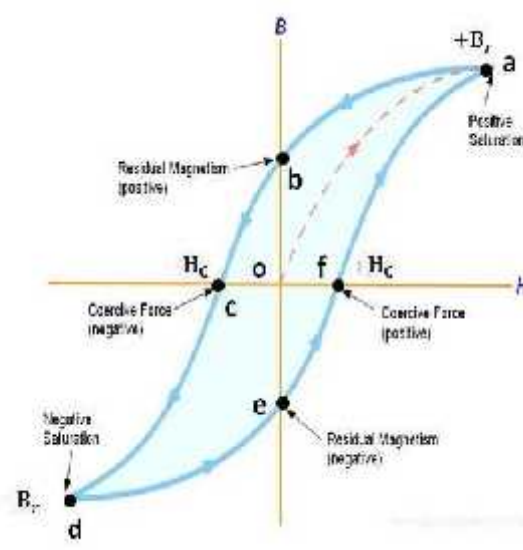


Fig: B-H Curve

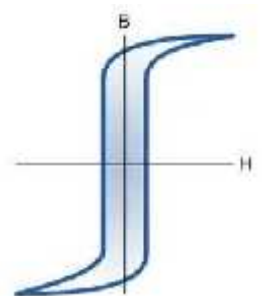
Based on the area of the hysteresis loop, the magnetic materials are classified into soft and hard magnetic materials.

8.11. Soft magnetic materials:-

The materials which can be easily magnetized and demagnetized are called Soft magnetic materials.

Properties:

- They can be easily magnetized and demagnetized and hence they show high values of susceptibility and permeability.
- Movement of domain walls is easy and hence even for small applied field large magnetization occurs.
- The nature of hysteresis loop is very narrow
- The hysteresis loop area is very small hence the hysteresis loss is also small as shown in fig.
- The coercivity and retentivity values are small
- These materials are free from irregularities or impurities or imperfections



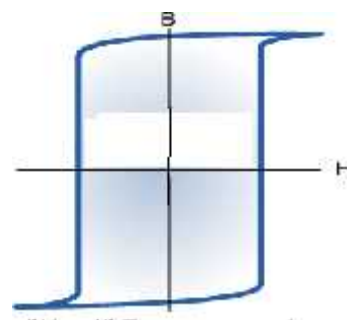
- Examples:
Fe-Sialloys, Ni-Fealloys, Fe-Coalloys, Ferrities and Garnet sets
- Applications:
 - They are used in switching devices, electromagnets,
 - They are used in matrix storage of computers.
 - They are used in motors, relays and sensors
 - They are used to make the temporary magnets.

Hard magnetic materials:-

The materials which can't be easily magnetized and demagnetized are called hard magnetic materials.

Properties:

- They can't be easily magnetized and demagnetized and hence they show low values of susceptibility and permeability.
- Movement of domain walls is not easy due to presence of impurities and hence large magnetic field is required for magnetization
- The nature of hysteresis loop is very broad.
- The hysteresis loop area is large hence the hysteresis loss is also large as shown in fig.
- The coercivity and retentivity values are high
- These materials have irregularities or impurities or imperfections
- Examples:
Carbon steel, tungsten steel, chromium steel, Cu-Ni-Fe alloys
Cu-Ni-Co alloys
Al-Ni-Co alloys
- Applications:
 - They are used in magnetic detectors,
 - They are used in microphones.
 - They are used in magnetic separators.
 - They are used to make the permanent magnets.



9. Practice Quiz

1. The dielectric constant $\epsilon_r =$
(a) ϵ_0/ϵ (b) ϵ/ϵ_0 (c) $\epsilon_0\epsilon$ (d) none of these
2. Dielectrics are:
(a) Metals (b) Semiconductors (c) Insulating materials (d) none
3. The spontaneous polarization depends on
(a) temperature (b) shape of the hysteresis
(c) dimension of the dielectric (d) all the above
4. A dielectric material can be polarised by applying ----- field on it
(a) gravitational (b) magnetic (c) electric (d) meson

5. Orientational polarisation is due to the ----- polar molecules in dielectric substance (a) change in separation (b) rotation (c) both a & b (d) none
6. The polarisation which is dependent on temperature is (a) electronic (b) Orientation (c) ionic (d) all the above
7. The orientation polarizability is (a) inversely proportional to T (b) directly proportional to T (c) independent of T (d) none of these
8. The product of charge and distance between charges is known as (a) dipole moment (b) dielectric constant (c) dipole (d) none of these
9. Which of the following substances is dielectric (a) Germanium (b) Copper (c) Mica (d) Tungsten
10. The electric susceptibility (a) P-E (b) P/E (c) PE (d) None of these
11. The Clausius-Mossotti equation does not hold for (a) liquids (b) gases (c) crystalline solids (d) amorphous solids
12. The Clausius-Mossotti relation is (a) $\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0}$ (b) $\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0}$ (c) $\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0}$ (d) $\frac{\epsilon_r - 1}{\epsilon_r + 2} = \frac{N\alpha}{3\epsilon_0}$
13. The electronic polarisation takes place in (a) ionic dielectrics (b) polar dielectrics (c) all the dielectrics (d) none of these
14. The crystal structure of BaTiO₃ is reduced to below its Curie temperature, the titanium and barium ions move up by 2.8% and oxygen ions move down by -% (a) 3 (b) 5 (c) 1 (d) 2
15. The crystal structure of BaTiO₃ above its Curie temperature is ----- (a) hexagonal (b) rhombohedral (c) cubic (d) triclinic
16. If 'p' is the polarisation of a dielectric with dielectric constant ' ϵ_r ' and E is the resultant electric field, then the internal field is given by (a) $E + \frac{P}{3\epsilon_0}$ (b) $E + \frac{P}{\epsilon_0}$ (c) $E + \frac{P}{\epsilon_0}$ (d) $3E + \frac{P}{\epsilon_0}$
17. Polarisation per unit electric field is known as (a) dielectric constant (b) dielectric susceptibility (c) dielectric polarisation (d) dielectric polarizability
18. The electronic polarisation shows remarkable variations in the following region (a) Microwave (b) Infrared (c) Ultraviolet (d) Variable
19. The unit for relative permittivity is (a) F/m (b) H/m (c) C/m (d) dimensionless

20. Hysteresis in polarization-electric field relation is exhibited by
(a) electro-optic materials (b) piezoelectric materials
(c) pyroelectric materials (d) ferroelectric materials
21. What will happen with magnetic material kept in external magnetic field?
a) They will move
b) They will develop magnetic lines of force
c) They will create a permanent magnetic moment
d) They will be unaffected
22. Magnetism originates due to rotational motion of charged particles. True or false?

Ans: True

23. What is the name of the continuous curve in magnetic field the tangent of which gives the direction of magnetic intensity?

- a) Magnetic lines of induction
b) Magnetic force
c) Magnetic lines of induction
d) Magnetic dipole moment
24. How is the residual magnetism from a material removed?

- a) Retentivity
b) Coercivity
c) Magneton
d) Switching off magnetic field

25. In which of the following magnetic moment is zero?

- a) Diamagnetic material
b) Paramagnetic material
c) Ferromagnetic material
d) Ferrimagnetic material

26. Which of the following is a weak magnet?

- a) Diamagnetic material
b) Paramagnetic material
c) Ferromagnetic material
d) Ferrimagnetic material



27. One Bohr magneton μ_B is equal to

- a) $\frac{4\pi m}{eh}$ (b) $4\pi mhe$
 (c) $\frac{me}{4\pi h}$ (d) $\frac{eh}{4\pi m}$

28. Magnetic susceptibility is negative for paramagnetic material. True or false?

Ans:

True 29. Unit of resistivity is

s:

- a) ohm-metre
 b) ohm/ metre
 c) ohm/metre²
 d) ohm/metre³

30. A magnetic field exists around

- a) Iron
 b) Copper
 c) Aluminium
 d) Moving charges

31. The unit of relative permeability is

- a) Henry/meter
 b) Henry/sq. meter
 c) Henry
 d) It is dimensionless

32. Which of the following type of materials are not very important for engineering application?

- a) Diamagnetic material
 b) Paramagnetic material
 c) Ferromagnetic material
 d) Ferrimagnetic material

33. The magnetic materials exhibit the property of magnetization because of

- a) Orbital motion of electrons
 b) Spin of electrons
 c) Spin of nucleus
 d) All of the above



34. For which of the following materials the net magnetic moment should be zero?

- a) Diamagnetic materials
- b) Ferrimagnetic materials
- c) Antiferromagnetic materials
- d) Antiferromagnetic

materials 35. The reciprocal of resistivity is

known as:

- a) susceptibility
- b) conductivity
- c) permeability
- d) permittivity

36. Which of the following statements is correct?

- a) The conductivity of ferrites is better than ferromagnetic materials
- b) The conductivity of ferromagnetic materials is better than ferrites
- c) The conductivity of ferrites is very high
- d) The conductivity of ferrites is same as that of ferromagnetic materials

37. Magnetism of a magnet can be destroyed by

- a) heating
- b) hammering
- c) by inductive action of another magnet
- d) by all above methods

38. Curie temperature is the temperature above which

- (a) a ferromagnetic material becomes paramagnetic.
- (b) a ferromagnetic material becomes diamagnetic.
- (c) a paramagnetic material becomes diamagnetic.
- (d) a paramagnetic material becomes ferromagnetic.

39. If the magnetising field on a ferromagnetic material is increased, its permeability.

- (a) is decreased
- (b) is increased
- (c) is unaffected
- (d) may be increased or decreased

10. Assignment Questions:

S.No	Question	BL	CO
1	Explain electronic polarization in a dielectric.	1	1
2	Derive an expression for Lorentz internal field in dielectrics.	1	1
3	What is ferro-electricity? Explain the properties of ferroelectrics.	3	1
4	Define orbital magnetic dipole moment. Calculate the orbital magnetic dipole moment for a revolving electron around the nucleus in an atom.	2	1
5	State and explain hysteresis behavior of ferromagnetic materials.	1	1
6	Distinguish between soft and hard magnetic materials.	2	1

11. Part A-Question & Answers

S.No		BL	CO
1	Define dielectric polarization. The process of producing electric dipoles by an electric field is called dielectric polarization in dielectrics.	1	1
2	Define dielectric polarizability. Answer: The induced dipole moment is proportional to the intensity of the electric field. $\vec{\mu} = a E$ Where a is the proportionality constant, called Polarizability	1	1
3	Define dielectric constant. Answer: It is defined as the ratio of permittivity of medium to permittivity of free space. $\epsilon_r = \frac{\epsilon}{\epsilon_0}$	1	1

4	What are ferro-electric materials? Give examples. Answer: Materials which exhibit electronic polarization even in the absence of the applied electrical field are known as ferro-electric materials. Example. Barium Titanate (BaTiO_3) Potassium Dihydrogen Phosphate (KH_2PO_4)	1	1
5	What is the relationship between relative permeability and magnetic susceptibility? Answer: When a magnetic material is magnetized by placing it in a magnetic field, the resultant field inside the material is the sum of the field due to the magnetization of the material and the original magnetizing field. The resultant field is called magnetic induction or magnetic flux density B. $B = \mu_0 H + \mu_0 M$ $= \mu_0 (H + M)$ $\mu H = \mu_0 (H + M) \quad (B = \mu H \text{ and } \mu = \mu_0 \mu_r)$ $\mu_0 \mu_r H = \mu_0 (1 + M) \frac{H}{H}$ $\mu_r = (1 + \chi_m)$ Where $\chi_m = \frac{M}{H}$. This is the relation between Relative Permeability and Magnetic Susceptibility.	1	1
6	Define Bohr magneton. Answer: Bohr magneton is a basic unit for orbital electron spins magnetic dipole moment. Where $\mu_B = \frac{eh}{4m\pi}$ is known as Bohr magneton and its value is $9.27 \times 10^{-24} \text{ A}\cdot\text{m}^2$.	1	1

12. Questions

S.No	Question	BL	CO
1	Describe the phenomena of electronic and ionic Polarization	1	1
2	Obtain an expression for the internal field (Local field) seen by an atom in an infinite array of atoms subjected to an external field.	1	1
3	Describe the origin of magnetic moment in an atom	1	1
4	Describe the classification of magnetic materials	2	1
5	Explain the hysteresis of ferromagnetic materials and explain soft and hard magnetic materials	2	1

13. Supportive Online Certification Courses

- Engineering Physics I (Experimental) by Prof. Alike Khare Prof. Pratima Agarwal Prof. S. Raviconducted by IIT Guwahati - 12 weeks
- Engineering Physics I (Theory) by Prof. Alike Khare Prof. Pratima Agarwal Prof. S. Raviconducted by IIT Guwahati - 12 weeks

14. Prescribed Text Books & Reference Books
Text Book

- A text book of Engineering Physics – Dr. M. N. Avadhanulu & Dr. P. G. Kshirsagar, S. Chand and Company, 11 Edition, 2019.
- Engineering Physics – B. K. Pandey and S. Chaturvedi, Cengage Learning, 2013.

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- Engineering Physics – K. Thyagarajan, McGraw Hill Publishers, 2019
- Engineering Physics - Sanjay D. Jain, D. Sahasrambudhe and Girish, University Press, 2010.
- Semiconductor physics and devices- Basic principle - Donald A. Neamen, McGraw Hill, 2011.

5. SolidStatePhysics,A.J.Dekker,MacmillanEducationUK,1969
6. Kittel's Introduction to Solid State Physics, Charles Kittel,WileyIndiaEditionPaperback,2019

